Problem Statement

PM Digest; Multi-Agent Orchestration; 05-pmgovernance-multiagent

The Problem / Opportunity

A Program Manager's actual workload; decisions to make, deadlines to track, blockers to unblock, people to follow up with; is scattered across three tools that don't talk to each other: Slack, Gmail, and Google Calendar. Checking each one separately every morning is slow, and it's easy to miss something that shows up in one channel but not another; a deadline mentioned in an email that's also the subject of an active Slack thread, for instance.

This project builds a multi-agent orchestration pipeline: a workflow that pulls information from multiple sources, routes each source through a specialized AI agent, and synthesizes everything into one prioritized, actionable digest delivered to a communication channel. The shape of the solution mirrors the cross-functional coordination that happens in any real program; decomposing an ambiguous problem into clear ownership (one agent per source), routing work to the right specialist, aggregating outputs, and keeping the whole thing reliable even when one piece fails or comes back empty.

Who's Affected

Program Managers and Technical Program Managers who need one trustworthy picture of what needs attention today; without personally checking three separate tools every morning, and without missing a cross-source connection that only becomes obvious once everything is read together.

Workflow Elements

- Scheduled trigger: firing reliably on a fixed cadence without manual intervention, and skipping cleanly; no duplicate work; if the pipeline has already run for the day.
- Central configuration + pre-flight validation: one source of truth for which sources are enabled, user context, and destination settings, with guardrails that catch missing configuration before agents run.
- Parallel multi-source ingestion: pulling live state from three independent systems of record (Slack, Gmail, Calendar) rather than working from stale or hardcoded data.
- Per-source specialized agents: each constrained to one data source, producing a consistent, structured output the next stage can consume.
- Cross-source synthesis: a dedicated agent that deduplicates overlapping items, resolves conflicts, prioritizes by urgency, and tags new vs. recurring context using prior-run history.
- End-to-end error handling: a failure anywhere in the chain surfacing as a visible alert, not a silent drop.

Each of those is a prerequisite to build the workflow.

What This Project Solves

- Ingests from three structurally different sources; a Slack channel search, a filtered Gmail query, and a Google Calendar feed; through one shared orchestration layer.
- Routes each source through a dedicated agent with a focused prompt and a defined input/output contract, so a change to one source's logic can't silently break another's.
- Merges and synthesizes all three outputs into a single digest that deduplicates cross-source overlap and flags VIP(Primary Stakeholders)-linked and time-sensitive items.
- Delivers the finished digest to Slack and logs it to Google Sheets as history, closing the loop so the next run can tell new items from recurring ones.
- Prevents duplicate work and duplicate output on repeated or manual runs via an early idempotency guard, rather than discarding duplicate results after the fact.